Credit Card Default Prediction: Comparing Classification Models and Decision Thresholds

Shi Xi (G2603572L) **Li Erteng (G2603157C)** **Jiang Chengrui (G2603177K)**
xi013@e.ntu.edu.sg erteng001@e.ntu.edu.sg chengrui001@e.ntu.edu.sg
Nanyang Technological University, MH6805 Machine Learning in Finance
github.com/ishixi233/CreditPrediction_KagSet_Course

Abstract

We build credit-risk classifiers for a card issuer and study how model ranking, decision thresholds and error costs interact. On the UCI Default of Credit Card Clients data (30,000 Taiwanese cardholders, 22.1% default) we compare six classifiers under one evaluation design: a stratified 70/30 split, five-fold cross-validation for configuration selection by average precision (AP), out-of-fold selection of the threshold that minimises a weighted error $r \cdot \mathrm{FN} + \mathrm{FP}$ for $r \in \{1, 2, 3, 5, 10\}$ with 5:1 as the pre-specified illustrative scenario, and a single evaluation of the frozen models on the test set, with calibration curves and Brier scores. Random Forest achieves the highest test AP (0.561; paired bootstrap intervals exclude every competitor) and the lowest cost for $r \geq 3$, but for $r \leq 2$ it is statistically indistinguishable from Logistic Regression, AdaBoost and a depth-5 Decision Tree. Threshold choice matters more than model choice: at the default threshold of 0.5 every model is more expensive than flagging everyone when $r = 5$, whereas cost-selected thresholds reduce cost by 23–29%. Gaussian Naive Bayes ranks acceptably (ROC-AUC 0.73) but its probabilities are worse than the base rate (Brier skill -0.67), showing why calibration must be checked before thresholds are read as probabilities.

1 Introduction

Banks score cardholders to decide whom to review, re-price or restrict. The score is probability-like, but the decision is binary and is taken by comparing the score with a threshold, so two kinds of error follow: a missed default (false negative, FN) and a false alarm on a paying customer (false positive, FP). Their costs differ and their ratio is rarely known. This separates two questions. *Which model ranks risk best?* is answered by threshold-free metrics such as average precision (AP) or ROC-AUC. *Which model and threshold should the bank operate?* is answered by the weighted error at one point on that ranking, and depends on the cost ratio and on calibration. The ranking winner need not be the operating winner [12].

Our proposal asked: *how do classification models and decision thresholds affect the trade-off between detecting defaults and generating false alarms?* Its objectives were to compare classifiers for next-month default under one framework, to evaluate them with metrics suited to imbalanced data, and to examine how thresholds trade missed defaults against false alarms under a stated cost scenario. Three design choices follow from the question. The 5:1 cost ratio taken from the German Credit data is treated as one scenario among five, so that conclusions are stated as a function of the ratio. Probability quality is assessed with calibration curves and the Brier score, because a threshold is only interpretable as a probability when the scores are calibrated. And the ranking metric and the operating cost are compared side by side, to show when they agree and when they do not.

Contributions. (i) A leakage-controlled comparison of six classifiers in which every selection decision is made inside the development portion and the test set is used once. (ii) A cost-sensitivity analysis over five FN:FP ratios with cost-minimising thresholds, costs relative to the best trivial policy and paired bootstrap intervals. (iii) A calibration analysis linking empirical thresholds to the theoretical $1/(1+r)$. (iv) An analysis of when the best-ranking model is and is not the best operating model. Code, splits, predictions, the frozen configuration and test outputs are in the repository.

Related work. The dataset was introduced by Yeh and Lien [14]. Credit-scoring benchmarks [9] find ensembles consistently strong but note that rankings do not translate directly into business value. Cost-sensitive decision theory [6] gives the optimal threshold of a calibrated scorer as $1/(1+r)$; Provost and Fawcett [12] show how iso-performance lines on the ROC plane select the operating point, so that different cost ratios can prefer different classifiers when ROC curves cross. Davis and Goadrich [4] motivate AP under imbalance; Niculescu-Mizil and Caruana [10] document the score compression of boosting and the overconfidence of naive Bayes, which we measure with the Brier score [2]. Selection bias from tuning on evaluation data [3] motivates our freeze-then-test protocol.

2 Data

We use the UCI Default of Credit Card Clients dataset [13]: 30,000 records, 23 predictors and a binary target (default on the payment of October 2005). Predictors are the credit limit, three demographic codes (sex, education, marital status), age, six monthly repayment-status codes (April–September 2005), six monthly bill amounts and six monthly payment amounts. The file has no missing values and no duplicate IDs; 6,636 clients (22.12%) defaulted.

Checks and recoding. Three features of the file required a decision. (1) `EDUCATION` contains the undocumented codes 0, 5 and 6 (345 rows) and `MARRIAGE` contains 0 (54 rows); we merge them into "other". (2) The repayment-status variables contain the undocumented codes -2 and 0, 58% of `PAY_0`; they are too frequent to be errors, so we keep them without assuming an ordinal meaning (Section 3). (3) 108 rows share identical predictors across 52 groups, 21 of which contain both labels, so they are distinct inactive accounts rather than duplicates; we keep them but never split a group across partitions or folds.

Split. A stratified 70/30 split with seed 42 gives a development set of 20,997 clients and a test set of 9,003 (default rates 22.13% and 22.10%). Five stratified folds inside the development set (4,195–4,204 clients each) are stored once and shared by all models. The test set is used once, after every configuration and threshold has been frozen.

3 Methods

3.1 Models and preprocessing

We compare Logistic Regression (LR), Gaussian Naive Bayes (GNB), k-nearest neighbours (KNN), a Decision Tree (DT), Random Forest (RF) [1] and AdaBoost [7] with decision stumps, all from scikit-learn [11], together with two baselines: the majority-class rule (never flag) and a constant score equal to the development default rate. The six models span linear, generative, distance-based, single-tree and ensemble approaches; Appendix F summarises each.

Preprocessing is the first step of every pipeline so that it is fitted only on the training part of each fold. `SEX`, `EDUCATION` and `MARRIAGE` are one-hot encoded. Numeric variables are standardised for LR, KNN and GNB and passed through unchanged for the tree models. Because the repayment codes $-2, -1, 0, 1, \ldots, 8$ are not ordinal, the encoding of the six `PAY` variables (numeric or one-hot) is a grid dimension for LR; one-hot raised its cross-validated AP from 0.503 to 0.544. Tree models split on the numeric codes directly.

Hyper-parameter grids. Grids were written before experiments and kept small (Appendix B): regularisation strength for LR ($C \in \{0.1, 1, 10\}$) crossed with the `PAY` encoding; k for KNN; maximum depth for DT ($\{3, 5, 8\}$) and RF ($\{5, 10, \infty\}$, 200 trees, five observations per leaf); learning rate for AdaBoost ($\{0.1, 0.5, 1\}$, 100 stumps). GNB has a single default configuration. The KNN grid

$\{11, 31, 61\}$ selected its boundary value and was extended once to $\{101, 151, 201\}$ before freezing; AP rises by only 0.001 from 101 to 201, so the selected $k = 201$ sits on a plateau.

3.2 Evaluation protocol

Ranking. Each configuration is scored by mean five-fold AP, computed from continuous scores for the default class; ROC-AUC is reported alongside. The configuration with the highest mean AP is selected per model, and the model with the highest mean AP is the primary model for discussion. Each development client also receives an out-of-fold (OOF) score from the pipeline fitted without its fold; these 20,997 scores per model are the input to the threshold and calibration analyses.

Thresholds and cost. A client is flagged when the score is at least a threshold t. For a cost ratio r (one missed default costs r false alarms), the weighted error is

$$C_r(t) = r \cdot \mathrm{FN}(t) + \mathrm{FP}(t), \qquad r \in \{1, 2, 3, 5, 10\}, \tag{1}$$

reported per client as $C_r(t)/N$. For each model and each r we choose $t^*_{m,r}$ to minimise C_r on the pooled OOF scores over every distinct decision from flagging no one to flagging everyone, breaking ties towards fewer false alarms. The ratio $r = 5$, taken from the cost matrix of the German Credit data [8], is the primary scenario; the default threshold 0.5 is the reference. Two trivial policies anchor every cost: flagging no one costs $r\pi$ per client and flagging everyone costs $1 - \pi$, where π is the default rate; they cross at $r = (1 - \pi)/\pi \approx 3.5$, so for $r \leq 3$ a model must beat "flag no one" and for $r \geq 5$ it must beat "flag everyone". We therefore report cost reduction relative to the better trivial policy, which makes columns comparable across r.

Calibration. For a calibrated scorer the cost-minimising threshold is $1/(1 + r)$ [6]; the distance between $t^*_{m,r}$ and this value is itself evidence of miscalibration. We report the Brier score, its skill relative to the constant baseline ($1 - \mathrm{Brier}/\mathrm{Brier}_{\mathrm{ref}}$, with $\mathrm{Brier}_{\mathrm{ref}} = \pi(1 - \pi) \approx 0.172$), and reliability diagrams with ten equal-frequency bins.

Freezing and testing. After selection, each pipeline is refitted on the full development set and its parameters and five thresholds are frozen. The test set is then scored once, and we report AP, ROC-AUC, Brier and, for every r, the confusion matrix, precision, recall, error rate and cost at the frozen and default thresholds. A paired stratified bootstrap of the test set (2,000 resamples, the same resample for all models) gives 95% intervals for each metric and for differences against the primary model [5]. The cost at the test-optimal threshold is also reported, to quantify the loss from transferring OOF thresholds.

4 Results

4.1 Ranking performance

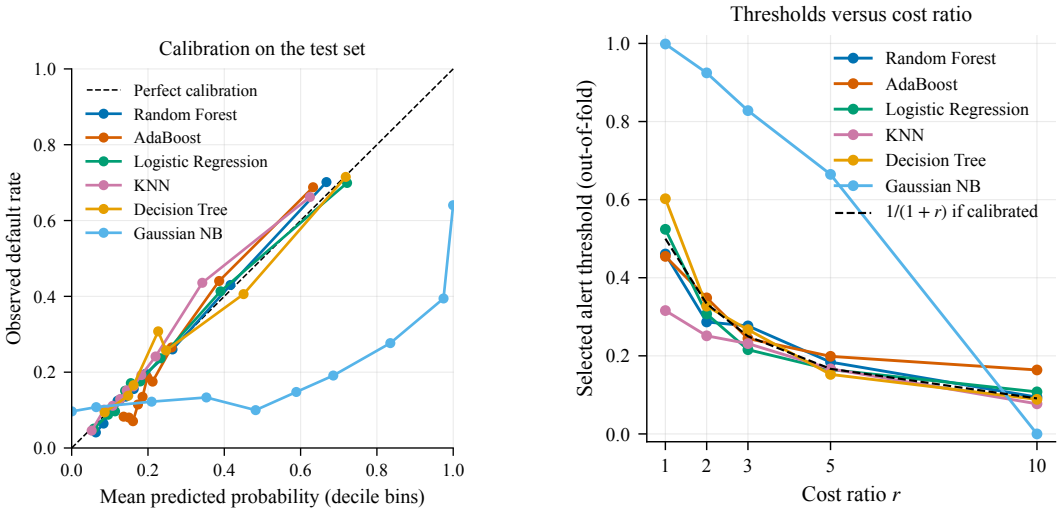
Table 1 reports cross-validated and test ranking metrics. Random Forest is best on every ranking metric; its test AP of 0.561 exceeds AdaBoost by 0.022 and the paired bootstrap interval for this difference (0.013–0.032) excludes zero, as do the intervals against every other model. AdaBoost and LR are indistinguishable (0.539 vs. 0.538). Cross-validated and test AP differ by at most 0.010. All models clear the constant baseline (AP 0.221) by a wide margin and the AP and ROC-AUC orderings agree.

4.2 Calibration

Figure 1a shows reliability diagrams on the test set. LR, RF and DT lie close to the diagonal across the whole range. AdaBoost's scores are compressed into $[0.1, 0.8]$: its lowest decile over-predicts (mean score 0.13 against an observed 0.07) and its highest under-predicts, as expected from boosting [10]; KNN slightly over-predicts in the upper deciles. GNB is the outlier: its scores pile up at 0 and 1 while the observed default rate in those bins is 0.10 and 0.64, giving a Brier score (0.288) far worse than the base-rate forecast (skill -0.67) despite a usable ranking (ROC-AUC 0.73). The Brier ordering thus differs from the AP ordering: DT is third by Brier but fifth by AP.

Table 1: Ranking and probability quality. Selected configurations are listed in Appendix B. CV: mean $\pm$ SD over five development folds. Test: single evaluation of the frozen model; AP interval from the paired bootstrap. Brier skill is relative to the constant baseline (0.172).

Model	CV AP	Test AP	95% interval	ROC-AUC	Brier	Skill
Random Forest	0.557 ± 0.006	**0.561**	0.540–0.584	**0.781**	**0.134**	**0.22**
AdaBoost	0.536 ± 0.004	0.539	0.518–0.562	0.765	0.139	0.19
Logistic Regression	0.544 ± 0.008	0.538	0.518–0.562	0.765	0.136	0.21
KNN	0.516 ± 0.007	0.521	0.501–0.543	0.759	0.141	0.18
Decision Tree	0.511 ± 0.008	0.508	0.488–0.529	0.751	0.138	0.20
Gaussian NB	0.487 ± 0.015	0.497	0.476–0.518	0.734	0.288	−0.67
Constant score	0.221	0.221	–	0.500	0.172	0.00



(a) Reliability diagrams (decile bins).

(b) OOF-selected thresholds against $1/(1+r)$.

Figure 1: Probability quality on the test set and its consequence for thresholds. Models whose scores are calibrated select thresholds close to the theoretical $1/(1+r)$; GNB selects 0.999 at $r = 1$ and 6×10^{-6} at $r = 10$.

4.3 Thresholds and cost sensitivity

Figure 1b shows how the cost-minimising threshold falls as r rises. For the five models with reasonable calibration the thresholds track $1/(1+r)$ (for RF: 0.46, 0.29, 0.28, 0.18, 0.09 against 0.50, 0.33, 0.25, 0.17, 0.09); GNB's thresholds are uninformative as probabilities. Table 2 gives the test cost per client at the frozen thresholds for every model and ratio, and Figure 2 the same numbers relative to the better trivial policy.

Three observations follow. At the primary ratio $r = 5$ Random Forest costs 0.553 per client, 29% below flagging everyone (0.779); the other models save 23–27%. At $r = 10$ the advantage shrinks to 0–11%: Decision Tree (0.792) is dearer than flagging everyone and GNB equals it, because at a threshold low enough to catch nearly all defaulters a coarse or miscalibrated score adds nothing. The default threshold 0.5 is costly: at $r = 5$ it gives 0.63–0.84 for every model, worse than the trivial policy, because it flags only 11–14% of clients and misses two thirds of defaulters (Figure 4); moving RF to 0.184 raises recall from 0.35 to 0.70 for 1,628 extra false alarms and cuts cost from 0.752 to 0.553.

4.4 Ranking winner versus operating winner

Table 3 places the AP ranking beside the cost ranking at each ratio. Random Forest, first by AP, is cheapest only for $r \geq 3$; at $r = 1$ the Decision Tree (fifth by AP) is nominally cheapest, at $r = 2$ Logistic Regression is, and AdaBoost, second by AP, is fourth by cost at $r = 1$ and $r = 10$. The

Table 2: Test cost per client, $C_r(t^*)/N$, at the thresholds frozen from out-of-fold predictions. Bold marks the lowest cost per column. The last rows are the trivial policies.

	Cost ratio r				
Model	1	2	3	5	10
Random Forest	0.179	0.307	**0.402**	**0.553**	**0.695**
AdaBoost	0.181	0.309	0.413	0.569	0.732
Logistic Regression	0.179	**0.306**	0.416	0.587	0.730
KNN	0.191	0.315	0.417	0.590	0.723
Decision Tree	**0.178**	0.313	0.431	0.591	0.792
Gaussian NB	0.193	0.332	0.431	0.599	0.779
Flag no one (majority class)	0.221	0.442	0.663	1.105	2.210
Flag everyone	0.779	0.779	0.779	0.779	0.779

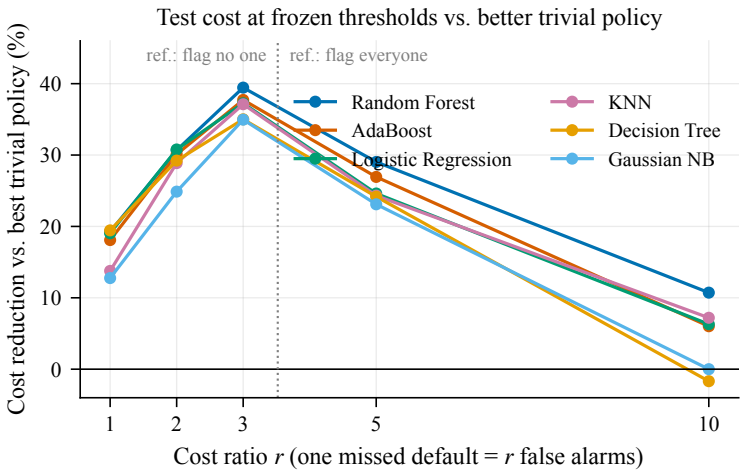


Figure 2: Cost reduction relative to the better trivial policy on the test set. Left of the dotted line the reference is "flag no one"; right of it, "flag everyone".

bootstrap decides which reversals are real. For $r \leq 2$ the cost differences between RF, AdaBoost, LR and DT are within ± 0.01 per client and every paired interval against RF contains zero, so the four models cannot be separated and the reversal is noise. For $r \geq 3$ every interval against RF excludes zero (at $r = 5$, RF vs. AdaBoost: 0.001–0.031), so RF's advantage is real where the costs are large. Figure 3 shows why: the three ROC curves almost coincide near the origin, where the $r = 1$ operating points lie, and separate in the middle, where $r = 5$ and $r = 10$ operate.

Transferring thresholds from out-of-fold predictions to the refitted model cost little: the gap between the frozen-threshold cost and the test-optimal cost is at most 0.013 per client (Decision Tree at $r = 10$) and below 0.004 for RF at every ratio except $r = 3$ (0.006).

4.5 What the models use

All four models that expose importances agree that the most recent repayment status, `PAY_0`, dominates (Appendix D): it carries 69% of the Decision Tree's impurity reduction, 48% of AdaBoost's and 25% of the Random Forest's, and the four largest LR coefficients are its categories (two months overdue: $+1.05$ log-odds; three months: $+0.98$; code 0: -1.24; code -2: -0.96). Earlier repayment codes and the two most recent payment amounts follow; AdaBoost leans more on payment amounts. The credit limit has a negative coefficient and ranks in the top eight for every tree. These are predictive, not causal, associations (Section 5).

Model	AP	r=1	2	3	5	10
Random Forest	1	2°	2°	1	1	1
AdaBoost	2	4°	3°	2	2	4
Logistic Regr.	3	3°	1°	3	3	3
KNN	4	5	5	4	4	2
Decision Tree	5	1°	4°	5	5	6
Gaussian NB	6	6	6	6	6	5

Table 3: Rank by test AP versus rank by test cost at each cost ratio (1 = best). °: cost difference from Random Forest has a bootstrap interval containing zero.

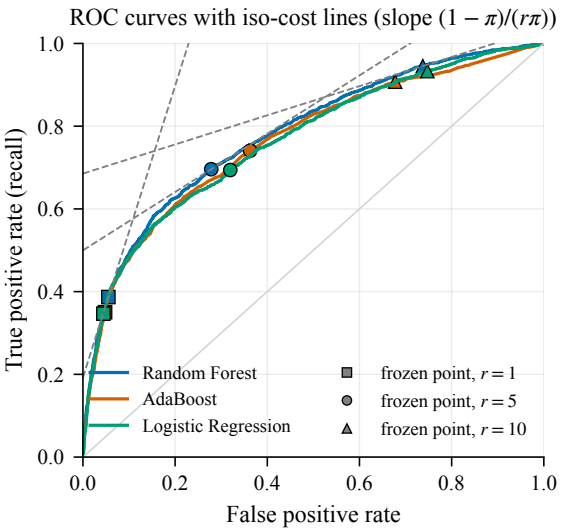


Figure 3: Test ROC curves of the three best-ranking models with frozen operating points for $r = 1, 5, 10$ and iso-cost lines through the Random Forest points.

5 Discussion

Why the best-ranking model need not minimise weighted error. AP integrates precision over all recall levels, whereas $C_r(t)$ depends on a single point of the ROC curve, the one where the curve's slope equals $(1-\pi)/(r\pi)$ [12]. A model can win the integral and lose the point in three ways, and we observe all three. (i) Curves that cross: near the origin the three best ROC curves coincide (Figure 3), so for $r \leq 2$ no model wins and a simpler one can be nominally cheapest; RF's advantage lies mid-curve, where $r \geq 3$ operates. (ii) Threshold transfer: t^* is estimated on OOF scores from five fold models and applied to a model refitted on all data; coarse scorers (DT has 109 distinct scores, KNN 162) are most exposed, and DT's regret of 0.013 at $r = 10$ is the largest. (iii) Noise: with 1,990 test defaulters, differences of 0.001–0.003 lie well inside bootstrap intervals, so the $r = 1$ "win" of the Decision Tree is not evidence. The conclusion is therefore conditional: for $r \geq 3$ use Random Forest; for $r \leq 2$ the four leading models are equivalent and simplicity favours Logistic Regression.

From results to a lending decision. Three qualifications separate "RF has the lowest cost" from "deploy RF at $t = 0.184$". The cost ratio is unknown: 5:1 comes from another dataset and country, so the recommendation is stated over a range, $3 \leq r \leq 10$, rather than at a point. As r grows, flagging everyone becomes competitive (the best model saves 11% at $r = 10$), so the question becomes how much review capacity the bank has rather than which classifier to use. Reading scores as probabilities requires calibration: RF, LR and DT qualify, AdaBoost's scores should be rescaled, and GNB's should not be read as probabilities even though its ranking is usable. The negative credit-limit coefficient is a selection effect, since limits are set by the bank in response to creditworthiness, not a protective effect of larger limits.

Limitations. The true cost ratio is unknown; the conclusion is bounded over r but not located. The data are one Taiwanese issuer in 2005 and the holdout is random, so performance in a later period or another market is unassessed. The sample contains only approved cardholders, so the models do not apply to applicants. Thresholds were selected on pooled OOF scores and transferred to refitted models, with a regret of at most 0.013. Grids were small, and $k = 201$ for KNN lies at the boundary of its grid. The repayment codes -2 and 0 lack an official definition, and impurity importances are not comparable across models.

Future work. Time-ordered validation on data with a longer history; post-hoc calibration (isotonic or Platt) of AdaBoost and GNB before thresholding; and a cost ratio elicited from loss-given-default and attrition data, which would turn the sensitivity analysis into a point recommendation.

6 Conclusion

Under one leakage-controlled protocol, Random Forest ranks credit-card defaulters best and, for cost ratios of 3:1 and above, also yields the lowest weighted error; for lower ratios it is tied with Logistic Regression, AdaBoost and a shallow Decision Tree. The larger effect is the threshold: operating at 0.5 makes every model worse than a trivial policy in the primary scenario, while out-of-fold thresholds for an explicit cost ratio cut cost by a quarter or more. Calibration decides whether those thresholds can be read as probabilities; Gaussian Naive Bayes shows that a usable ranking can coexist with unusable probabilities.

Member contributions and work plan

Shi Xi: Logistic Regression and Gaussian Naive Bayes; data checks, preprocessing and financial interpretation (proposed); in addition, the shared split and evaluation protocol, the cross-validation and freeze-then-test pipeline, calibration and uncertainty analyses, feature interpretation, and drafting and integration of the report (delivered). **Li Erteng**: KNN and Decision Tree; shared split, cross-validation and experiment records (proposed); verification of the frozen configuration, the experimental-design section, the rank-comparison analysis and the corresponding slides (delivered). **Jiang Chengrui**: Random Forest and AdaBoost; results visualisation and threshold analysis (proposed); threshold optimisation and cost-sensitivity analysis over five ratios, out-of-fold cost figures, the results section on thresholds and the corresponding slides (delivered). All members reviewed the shared pipeline and contributed to the report and slides. Overall contributions are approximately **40% (Shi Xi), 30% (Li Erteng), 30% (Jiang Chengrui)**.

References

[1] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[2] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78 (1):1–3, 1950.

[3] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

[4] Jesse Davis and Mark Goadrich. The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, pages 233–240, 2006.

[5] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.

[6] Charles Elkan. The foundations of cost-sensitive learning. In *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 973–978, 2001.

[7] Yoav Freund and Robert E. Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1):119–139, 1997.

[8] Hans Hofmann. Statlog (German credit data). UCI Machine Learning Repository, 1994.

[9] Stefan Lessmann, Bart Baesens, Hsin-Vonn Seow, and Lyn C. Thomas. Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research*, 247(1):124–136, 2015.

[10] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005.

[11] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[12] Foster Provost and Tom Fawcett. Robust classification for imprecise environments. *Machine Learning*, 42 (3):203–231, 2001.

[13] I-Cheng Yeh. Default of credit card clients. UCI Machine Learning Repository, 2009.

[14] I-Cheng Yeh and Che-hui Lien. The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Systems with Applications*, 36(2):2473–2480, 2009.

A Data dictionary and recoding

Amounts are in New Taiwan dollars. Monthly variables are indexed from the most recent month (September 2005) backwards; the repayment-status columns are named `PAY_0`, `PAY_2`, ..., `PAY_6` in the source file (there is no `PAY_1`) and we keep the original names.

Table 4: Variables, documented codes, undocumented codes found in the file, and the preprocessing applied.

Variable	Meaning	Values	Preprocessing
`ID`	Client identifier	1–30,000, unique	Excluded from predictors; used to align splits and predictions
`LIMIT_BAL`	Credit limit (individual and family)	10,000–1,000,000	Numeric
`SEX`	Sex	1 male, 2 female	One-hot
`EDUCATION`	Education	1 graduate school, 2 university, 3 high school, 4 others; undocumented 0 (14 rows), 5 (280), 6 (51)	0, 5, 6 merged into 4 (345 rows), then one-hot
`MARRIAGE`	Marital status	1 married, 2 single, 3 others; undocumented 0 (54 rows)	0 merged into 3, then one-hot
`AGE`	Age in years	21–79	Numeric
`PAY_0`, `PAY_2`–`PAY_6`	Repayment status, September to April 2005	Documented: -1 paid duly, 1–8 months of delay, 9 nine months or more; undocumented -2 and 0 (together 58% of `PAY_0`)	Numeric for tree models and GNB; one-hot for LR (selected by cross-validation)
`BILL_AMT1`–`BILL_AMT6`	Bill statement amount, September to April	May be negative (overpayment)	Numeric
`PAY_AMT1`–`PAY_AMT6`	Amount paid, September to April	≥ 0	Numeric
Target	Default on the October 2005 payment	1 default (6,636), 0 otherwise (23,364)	Label

The file has no missing values, so no imputation step exists. Numeric variables are standardised only for LR, KNN and GNB. The 108 rows with identical predictors (52 groups, 21 containing both labels) are kept and never split across partitions or folds.

B Hyper-parameter grids and selected configurations

Table 5: Pre-stated grids, selected configuration and mean CV AP per candidate. Simpler candidates are listed first and win ties.

Model	Grid (mean CV AP)	Selected
Logistic Regression	$C \in \{0.1, 1, 10\} \times$ `PAY` numeric (0.503, 0.503, 0.503) or one-hot (0.543, 0.544, 0.542)	$C = 1$, one-hot
Gaussian NB	default; `PAY` one-hot tried (0.491 vs. 0.487, within fold SD) and not adopted	default
KNN	$k \in \{11, 31, 61\}$ (0.462, 0.500, 0.512); extended once to $\{101, 151, 201\}$ (0.515, 0.516, 0.516)	$k = 201$
Decision Tree	depth $\in \{3, 5, 8\}$ (0.473, 0.511, 0.484)	depth 5
Random Forest	200 trees, min. 5 per leaf, depth $\in \{5, 10, \infty\}$ (0.551, 0.557, 0.552)	depth 10
AdaBoost	100 stumps, rate $\in \{0.1, 0.5, 1\}$ (0.536, 0.536, 0.535)	rate 0.1

Table 6: Test-set confusion counts and derived metrics at the frozen threshold for $r = 5$ ($N = 9{,}003$, 1,990 defaulters).

Model	t^*	TP	FP	FN	TN	Precision	Recall	Error rate	Cost/client
Random Forest	0.184	1385	1952	605	5061	0.415	0.696	0.284	0.553
AdaBoost	0.199	1474	2544	516	4469	0.367	0.741	0.340	0.569
Logistic Regression	0.165	1381	2242	609	4771	0.381	0.694	0.317	0.587
KNN	0.167	1404	2378	586	4635	0.371	0.706	0.329	0.590
Decision Tree	0.152	1530	3018	460	3995	0.336	0.769	0.386	0.591
Gaussian NB	0.665	1316	2023	674	4990	0.394	0.661	0.300	0.599

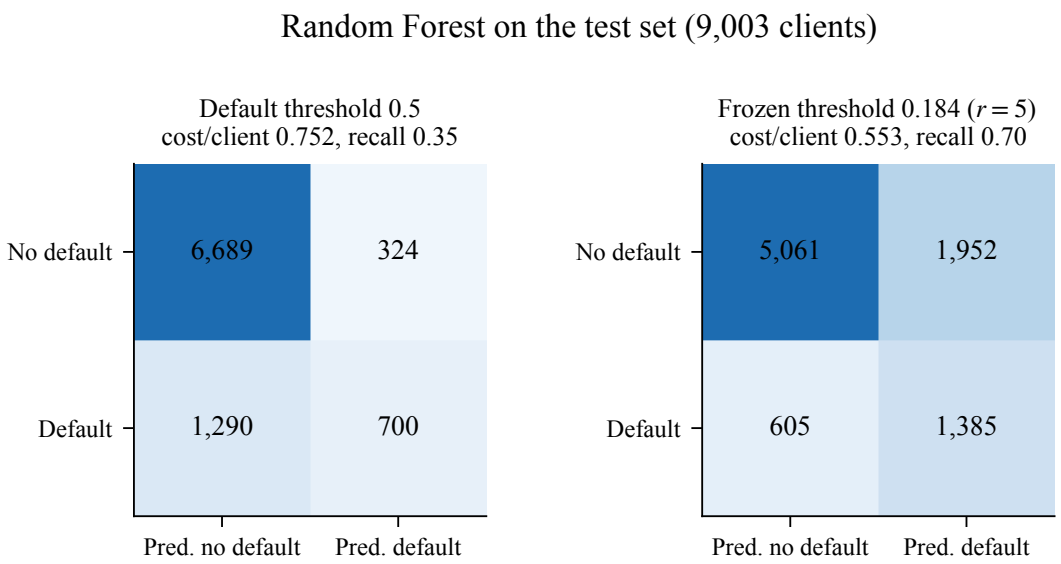


Figure 4: Random Forest confusion matrices on the test set at the default and the frozen ($r = 5$) threshold.

C Decision metrics at the primary scenario

D Feature interpretation

E Additional figures

F Model summaries

Logistic Regression models the log-odds of default as a linear function of the (standardised and one-hot) features and is fitted by penalised maximum likelihood; the tuned quantity is the inverse L2 strength C. Its scores are probabilities by construction and were the best calibrated in our study. **Gaussian Naive Bayes** applies Bayes' rule under conditional independence with a Gaussian likelihood per feature and class; it has no tuned hyper-parameter. The independence and normality assumptions are violated by correlated monthly amounts and discrete codes, which produces extreme posterior probabilities but leaves the ranking usable. **k-nearest neighbours** scores a client by the default rate among its k closest clients in standardised Euclidean distance; larger k smooths the estimate, and the scores take at most $k + 1$ distinct values. **Decision Tree** partitions the feature space by greedy impurity-minimising splits up to a maximum depth; the score is the default rate of the leaf, so depth 5 yields few distinct scores and a step-shaped ROC curve. **Random Forest** averages 200 trees, each grown on a bootstrap sample with a random subset of features at every split; averaging reduces variance and gives nearly continuous, well-calibrated scores. Depth and a minimum of five observations per leaf limit over-fitting. **AdaBoost** fits 100 depth-1 trees sequentially, re-weighting

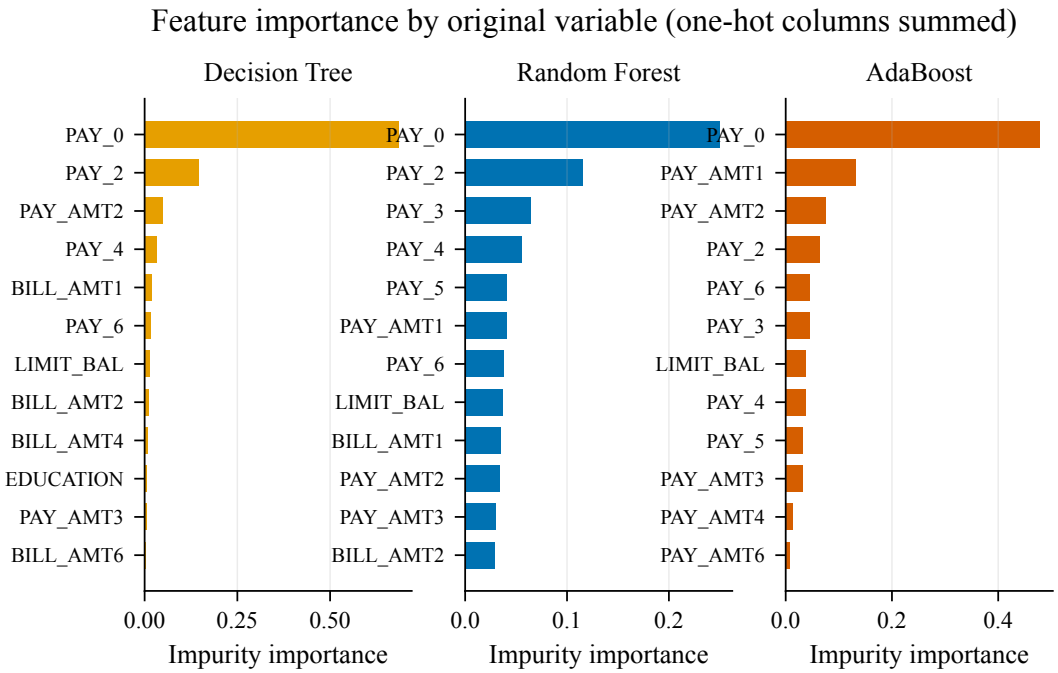


Figure 5: Impurity-based importance by original variable (one-hot columns summed) for the three tree models fitted on the full development set.

misclassified clients at each round; the score is a logistic transform of the weighted vote, which is known to be compressed towards 0.5, matching the calibration curve observed.

G Reproducibility

The repository contains the raw file (SHA-256 recorded), the split indices, every out-of-fold and test prediction, the frozen configuration with its commit hash, the fitted pipelines and the scripts `01`–`10` that regenerate every table and figure in this report. Package versions are pinned (`scikit-learn` 1.9.1, `numpy` 2.5.4). `python -m pytest` checks the split and the metric functions.

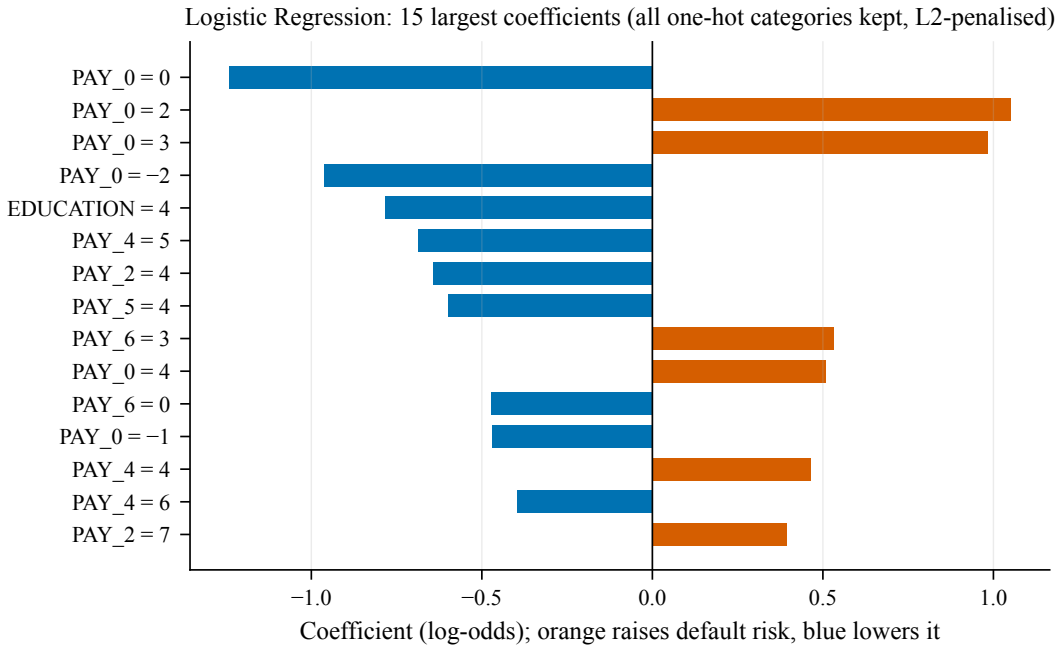
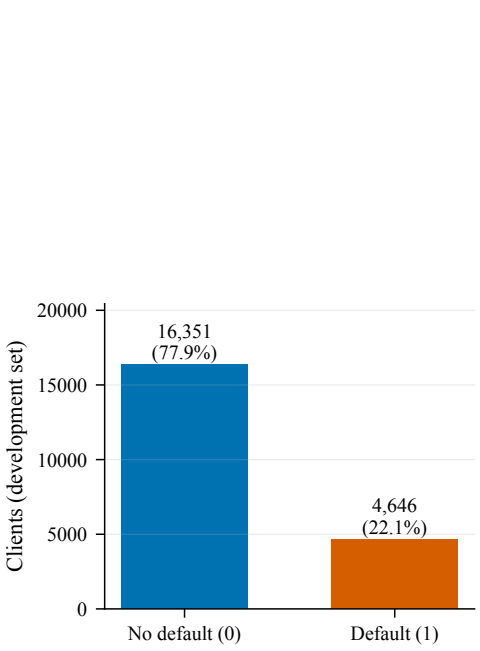
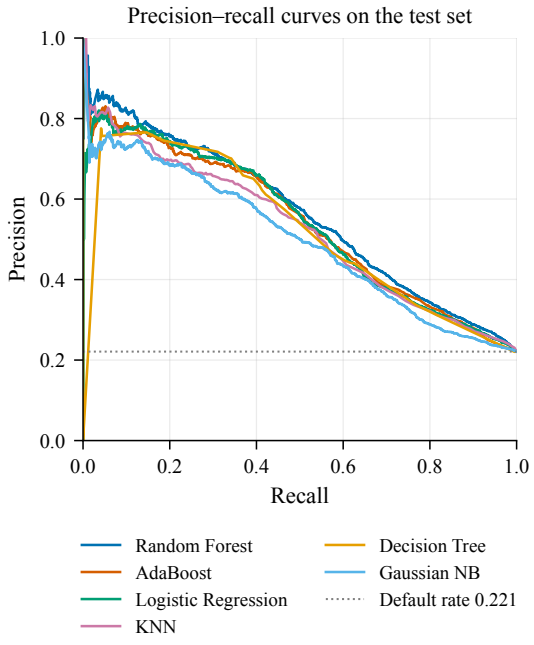


Figure 6: Fifteen largest Logistic Regression coefficients (all one-hot categories retained under L2 regularisation).



(a) Class balance in the development set.



(b) Precision–recall curves on the test set.

Figure 7: Supporting figures.